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RESEARCH ARTICLE

HARMONY: A Framework for Multimodal LLM-Powered AI Agents in Smart Homes via the Model Context Protocol

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ABSTRACT Multimodal Large Language Models (MLLMs) handle visual, audio, and text inputs effectively, enabling more natural interactions in applications like smart homes. Platforms such as Samsung SmartThings coordinate devices through open APIs, but integrating MLLMs into smart home ecosystems poses challenges including disjointed data sources, vendor-specific limits, and rule-based reactivity rather than true context awareness. This paper introduces a framework Home Autonomous Resilient Multimodal Operations Nexus for You (HARMONY) that combines multimodal generative Artificial Intelligence (AI) with standardized communication protocols for smart home ecosystems in a simulated home environment. HARMONY uses a central MLLM to break down user requests into device-specific actions and coordinate AI agents that connect to Samsung SmartThings via RESTful APIs. It employs Anthropic's Model Context Protocol (MCP) for standardized orchestration, enabling state queries, event subscriptions, data retrieval, action invocation, and consistent state management to perform relevant tasks in comfort, security, and culinary assistance. This setup delivers intuitive multimodal control while inferring user intent from diverse contextual signals. Through simulation-based evaluation, the study demonstrates a significant evolution in Large Language Models (LLM) orchestration capabilities from basic single-task models to context-reflective agents enhancing personalization, operational efficiency, and decision-making while highlighting the trade-offs between execution speed, semantic reasoning, and agentic flexibility.

INDEX TERMS Multimodal LLM, smart home, AI agents, model context protocol, human-computer interaction, Internet of Things (IoT), ambient intelligence.

I. INTRODUCTION

A smart home is characterized by the proliferation of interconnected devices and systems intended to automate and improve domestic life. It has evolved from a futuristic notion to a rapidly growing reality [1]. Smart environments, in the context of ubiquitous computing, integrate networks of sensors and devices into elements like wearables, appliances, light bulbs, and the built environment, enabling responsive interactions that recognize and adapt to user needs [2]. The Internet of Things (IoT) has driven substantial progress in smart home development, enabling sensors, software, and internet connectivity to collect and exchange data, forming complex home ecosystems [3], [4]. Smart devices are increasingly prevalent, though many systems still rely on predetermined rules or simple instructions, while newer AI-enhanced ecosystems incorporate predictive and contextual elements [5], [6], [7], [8], [9], [10]. Context-awareness implies that the system can collect, interpret, and utilize information regarding the state at hand, such as the physical situation (e.g., temperature, illumination), user state (e.g., location, activity), device status, interaction history, and even

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the implicit goals of the user to adapt its behavior [11]. Multimodal generative AI enhances smart home technology by allowing devices to process visual, textual, and audio inputs simultaneously, creating more intuitive and responsive environments [12]. This advances the long-standing vision of smart homes that sense needs, maximize comfort, and simplify daily routines [13]. Although tremendous progress has been made with a growing number of internet-enabled devices, current smart home ecosystems are challenged by the following [14]:

- **Siloed Interactions:** Devices from different brands or product lines often lack seamless interoperability, requiring proprietary hubs or manual bridging despite emerging open-source smart home connectivity standard like Matter.
- **Restricted Input Modalities:** Interactions remain dominated by voice commands (e.g., "Turn on the lights") or app controls, limiting richer multimodal human-like exchanges via vision or gesture.
- **Reactive Systems:** Most setups respond to explicit user instructions or basic triggers rather than proactively anticipating needs through advanced context.
- **Deep Contextual Understanding:** Systems process isolated sensor data and simple cues but struggle with holistic inference of user goals, history, and environmental states for truly adaptive decisions.

The emergence of MLLMs, such as Google's Gemini and OpenAI's GPT-4V, marks a paradigm shift in Artificial Intelligence (AI). These models process mixed visual and textual inputs—and, in integrated systems, audio—providing early indications that more natural and intelligent human–computer interfaces [15]. MLLMs extend traditional LLMs by incorporating multimodal input and output capabilities, enabling them to accept, reason over, and generate responses grounded in multiple data modalities. Prior to the development of MLLMs, an extensive research on multimodality is explored, which can broadly be categorized into discriminative [16], [17] and generative [18], [19] paradigms. Contrastive Language–Image Pre-training (CLIP) [16], an example of the former, maps visual and textual information into a shared representation space, supporting a wide range of downstream multimodal tasks.

Samsung, with its expansive and diverse smart device portfolio—from AI-enabled Bespoke appliances to QLED televisions and Galaxy mobile devices—offers a compelling real-world environment for exploring the full potential of MLLMs in smart home settings. Integrating Samsung's SmartThings ecosystem with MLLM APIs and emerging standards like Matter could represent a significant step toward scalable, context-aware, and multimodal intelligent home systems.

The ideation behind this proposed framework is based on observed research gaps in existing AI-driven smart home systems. Current smart home platforms rely heavily on device-specific APIs and rule-based automation, constraining their scalability, interoperability, and ability to support richer forms of context awareness. Open-source smart home connectivity standard such as Matter, have advanced cross-vendor compatibility, yet the research gaps exist in seamless coordination and intelligent adaptive behavior across heterogeneous devices. A standardized semantic communication layer, such as Anthropic's Model Context Protocol (MCP), is essential to enable plug-and-play orchestration, consistent state management, and structured device interaction. In addition, traditional AI approaches used in smart homes have reliably supported single-modality processing and reactive automation, but requires the dynamic, multimodal capabilities for advanced interactions. MLLMs facilitate the multimodal capability by enabling parallel processing of textual, audio, and visual inputs, providing more natural and context-sensitive interactions. Future intelligent home systems are expected to shift toward proactive, goal-directed reasoning to infer user intent, resolve ambiguity, and enable multi-agent coordination. Semantic comprehension and autonomy can benefit from AI agents orchestrated by a multimodal LLM, which remain largely experimental.

This work proposes HARMONY, a framework integrating multimodal AI, agent-based orchestration, and standardized device control via MCP and existing smart home APIs.The significant contributions of the proposed work are within a fully simulated smart home environment.

A. HARMONY FRAMEWORK DESIGN

A multimodal smart home orchestration framework that fuses MLLMs with custom, domain specific AI agents, leveraging Anthropic's established MCP for tool interactions. HARMONY employs dedicated agents for comfort and ambiance (CAA), security and safety (SSA), kitchen and culinary assistance (KCA), and entertainment and wellbeing (EWA), coordinated by a central Orchestrator Agent (OA). The MLLM processing core performs multimodal intent recognition, context fusion, and task planning, enabling intuitive, context-aware smart home control across ambiance, security, and cooking domains. By integrating the reasoning capabilities of MLLMs, the directed expertise of AI agents, the dense sensory information from Samsung devices, and the structured communication that comes from MCP, HARMONY establishes a simulation-based smart home experience that is:

- **Intuitively Interactive:** Responding to natural, multimodal human cues.
- **Contextually Aware:** Knowing the state of the environment and user behaviors.
- **Helpful Proactively:** Anticipating and performing acts of help or making suggestions.
- **Seamlessly Integrated:** Having multiple Samsung devices blended into seamless experiences.

B. SCALABLE DEVICE INTEGRATION

Adapts MCP protocol for the smart home device orchestration, creating a bidirectional interface between Samsung smart devices and AI agents between Samsung smart devices and AI agents, addressing the integration challenge and enabling real-time, multimodal control across heterogeneous environments.

C. ADVANCEMENT IN LLM ORCHESTRATION

Examines the evolution of LLM abilities—linear executors (e.g., GPT-4o Mini) to superior, context-sensitive agents (e.g., Gemini 2.5 Flash)—and initial explorations of LLM orchestration suggest trade-offs that could affect future smart home agent autonomy.

This work adapts existing technologies like Anthropic's MCP Protocol for smart home tool interactions, Samsung's SmartThings ecosystem with added orchestration, and commercial MLLMs (GPT-4, Gemini) via novel agent dispatch. It contributes novel elements:

(i) a five-agent hierarchical architecture (CAA, SSA, KCA, EWA, Orchestrator) enabling domain-specific task allocation,

(ii) Multimodal task decomposition workflow parsing compound prompts into agent tasks for smart home scenarios, and

(iii) The LLM Archetype Matrix, for systematically evaluating LLMs for orchestration suitability.

The rest of this paper is structured as follows: Section II reviews related work. Section III details the proposed architectural framework and workflow. Section-IV details the methods used, including core technologies and simulation setup, and presents a case study, including the simulation environment, agent configuration, and scenario analysis. Section V discusses the limitations of the proposed work, and outlines its directions for future research for handling the limitations specified. Finally, Section VI concludes the paper by summarizing the findings.

II. RELATED WORKS

Recent research have focused on enhancing the usability and flexibility of smart home systems through machine learning. Activity-based automation of domestic household work has been explored by Manu et al. [20] using deep learning algorithms to analyze user activity through accelerometer data, showing how sensor-driven inference can facilitate automated household routines. Building on these advances in activity recognition, voice-operated home assistants offer an intuitive interface for smart home control, enabling the system to comprehend verbal commands and act accordingly. Rani et al. [21] proposed a voice-controlled home automation system based on IoT and NLP technologies that can control common household appliances. Higher-level NLP algorithms are employed by commercially available intelligent assistants like Bixby, Google Assistant, and Alexa to enable a user-friendly experience. From shopping and setting reminders to controlling devices and automating household tasks, these systems are capable of managing a wide range of commands and queries. However, deciphering and reacting to implicit or vague user commands remains a frequent challenge for modern home assistants [22], [23], [24].

The literature [20], [21], [22], [23], [24] introduced task-specific language frameworks that constrain and structure user interaction. For example, Noura et al. [25] presented the VISH grammar and dataset for modelling smart home utterances and enhancing the natural language models' understanding of user objectives. By introducing a goal-oriented language, the Noura et al. approach [25] enables commands such as 'the blinds are too low' to be interpreted as explicit device actions. However, these systems envisages mechanisms for persistent goals or deeper reasoning about sensor data and multi-device coordination.

LLMs introduce promising new capabilities for smart home systems, particularly in enhancing natural language understanding and interaction, although their integration into commercial environments remains in early stages. LLMs are extensive language models that have been trained on a variety of corpora, including a sizable amount of the textual and coding content [26], [27]. These models stand out for their remarkable performance on a variety of tasks, all without necessitating major model adjustments for specific application cases [28], [29]. One well-known LLM, GPT-3, for example, has been used in a variety of contexts, including social dynamics modelling in natural language control of robots [30], [31] and in small-community setting [32].

Recently, LLMs have shown promise in enhancing user interaction in smart home environments. Recent systems like Sasha [33] utilize LLMs to target relevant devices and align with user preferences. SAGE functions as a semi-autonomous LLM-driven agent designed to provide more natural, human-like interaction within smart home environments. However, these systems primarily focus on text-based interactions, which can be challenging to capture user intent. Several contexts have investigated the use of diverse input and output modalities that are suited to user preferences and contextual requirements [34], [35], [36]. Several frameworks and architectures have been used to investigate this idea, showing that it is feasible in smart environments, such as smart houses [37]. As an example, systems combining touch, speech, and eye-blink inputs have been developed to enable users with limited physical mobility to control smart home devices [38].

Recent work on MLLMs builds upon these advances by expanding the range of input and output modalities [39]. Beyond text and images, researchers have extended MLLMs to accommodate data to generate outputs across modalities, including speech [40], video [41], and images [42]. NExT-GPT [43], which incorporates diffusion models, demonstrates this capability in a research setting by providing a framework that supports mixed-modality inputs and outputs—including combinations of text, images, audio, and video. Many commercial multimodal assistants still exhibit limited flexibility, often presenting information in similar formats across users and contexts despite growing personalization

FIGURE 1. HARMONY system architecture – integration of multimodal LLM core (Qwen 2.5 VL), specialized AI agents, and samsung smart devices via the model context protocol (MCP) for context-aware smart home orchestration.

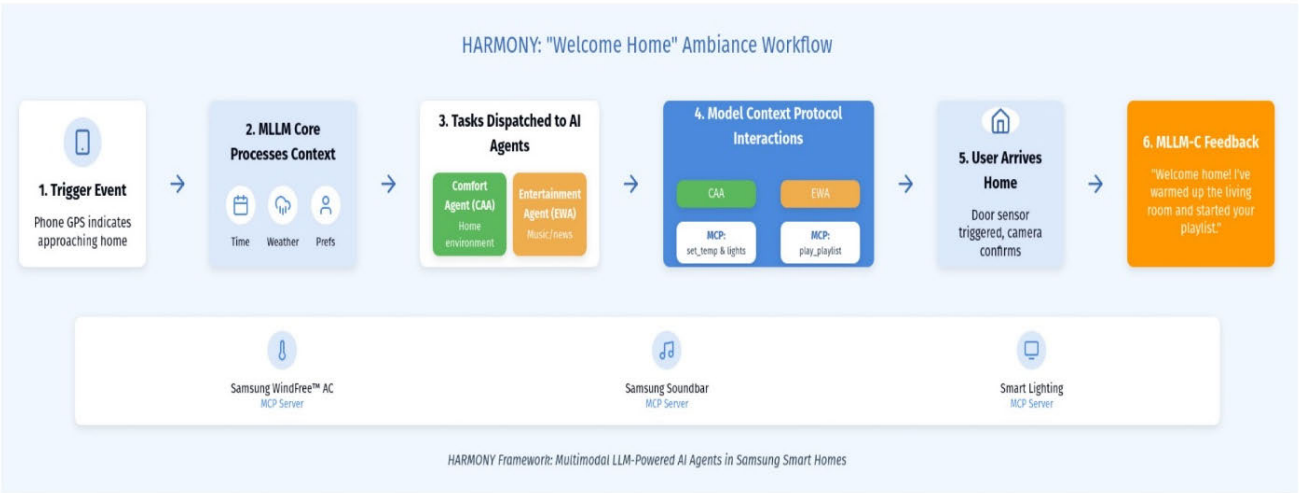

FIGURE 2. HARMONY "welcome home" ambiance workflow – multimodal trigger-to-execution pipeline for personalized smart home automation.

features. Adaptive interaction technologies have been developed to tailor system behavior to individual users and environments. Information from dedicated home sensors or embedded sensors in interaction devices is used to achieve this kind of adaptation [38]. Adapting to user preferences and environmental context has proven crucial in improving the interaction experience in smart home contexts [10], [44], [45], [46], [47], [48], [49].

From the literature [10], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], the following research gaps are observed:

(1) While machine learning, voice assistants, and IoT-NLP integrations have significantly advanced smart home automation, current systems poses challenges to comprehend commands that are unclear, implicit, or context-dependent. In addition, activity-based automation and voice-controlled systems have been limited in flexibility and generalizability.

(2) Current smart home systems face silos, text/voice limitations, and reactive control, despite NLP/LLM progress. Samsung's SmartThings AI Home integrates Bespoke appliances for multimodal (voice/vision/sensor), context-aware automation across domains, learning user patterns for proactive coordination without protocols like MCP.

(3) The fragmented ecosystem integration and the absence of continuous, goal-oriented orchestration currently limit the practical deployment of MLLMs in smart homes, despite their potential to accommodate various modalities and complicated output types (such as text, audio, and video).

To address the research gaps, the proposed HARMONY framework integrates MLMMs with specialized AI agents using SmartThings REST APIs. HARMONY employs an MLLM-driven processing core for multimodal intent recognition, context fusion, and task planning, while a dedicated Orchestrator Agent coordinates distributed domain agents (e.g., cooking, security, and ambiance). These agents execute

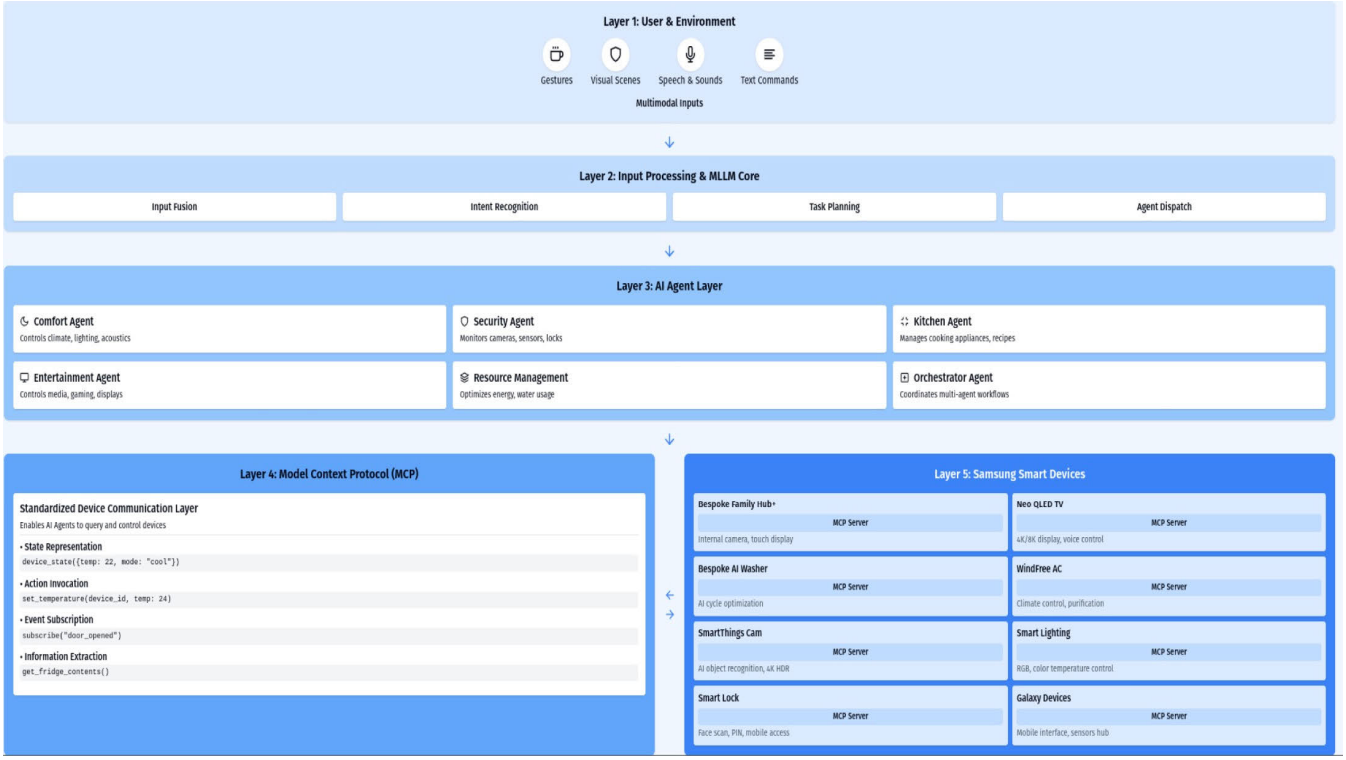

FIGURE 3. HARMONY layered system architecture – end-to-end multimodal orchestration for smart homes.

proactive, context-aware actions across Samsung's smart home ecosystem via structured MCP-based tool interactions.

III. PROPOSED ARCHITECTURE

HARMONY is a simulated smart home AI framework that leverages multimodal generative AI to create intelligent, context-aware interactions through synchronized multiple concurrent inputs—viz. Visual (camera streams), textual (typed inputs), and audio (voice commands)—and generate context-aware outputs. In contrast with legacy AI models that react to individual inputs, these systems work across synchronized flows of data, providing a context-sensitive, composite response mechanism.

In real-world scenario, the system allows a smart home to recognize an individual entering the kitchen (visual), hear them ask, "What can I cook tonight?" (audio), identify ingredients on the counter (visual), and show a recipe suggestion on the fridge or TV (textual/visual output), all seamlessly in one stream.

This model of high-fidelity interaction is the foundation of next-generation user experience (UX) in ambient intelligence. Fig. 1 and 2 depict the HARMONY framework, a multimodal LLM-driven architecture for intelligent home settings that follows a client-server model to enable modular communication between AI agents and IoT devices. Herein, user inputs-visual, audio, and text-are managed by a central MLLM core that delegates the tasks to specialized AI agents responsible for Comfort & Ambiance, Security, Kitchen, Entertainment, and orchestrator with maintenance and resource monitoring implemented as auxiliary sub-functions. These agents communicate with Samsung smart devices (appliances, displays, security systems, lighting, and mobile hubs) through the SmartThings REST API, which abstracts hardware heterogeneity. The system supports multimodal models such as Qwen 2.5 VL for context-aware interactions, coordinated by a centralized Orchestrator Agent that manages multi-agent workflows and ensures seamless orchestration of complex smart home tasks.

Fig. 3 illustrates the workflow on how a GPS-based trigger initiates multimodal context processing, and dispatching tasks to specialized AI agents (e.g., comfort and entertainment). These agents invoke device-level actions through the SmartThings REST API, resulting in a seamless, adaptive experience—such as preconditioning the environment and playing music—before the user physically arrives home. The process starts with a trigger event, where the smartphone GPS information of the user indicates proximity to the home. This geospatial input serves as a situational initiator, which is sensed and propagated to the MLLM Core. The MLLM core translates the event based on temporal information (time), ambient conditions, and user preferences, semantically processed to create high-level task directives executed by specialized AI agents.

Two of the most important agents are invoked within this workflow: the Comfort Agent (CAA), which is in charge of controlling environmental factors like temperature and

lighting, and the Entertainment Agent (EWA), whose task is to start media play based on the user's preferences. The agents communicate with the intelligent device infrastructure through MCP and facilitate controlled interactions by allowing supported commands such as state representation, action invocation, event subscriptions, and extraction of information. In particular, the CAA sends commands such as *"set_temp"* and "*lights_on*", whereas the EWA plays a playlist, with the two agents communicating with Samsung smart devices like the WindFree™AC, Samsung Soundbar, and Smart Lighting, all of which are MCP servers making device functionality available through lightweight APIs.

Once the user physically arrives—verified via door sensor activation and camera confirmation—the system transitions to a confirmation phase. The MLLM generates natural language feedback based on the execution status of the invoked commands and presents it to the user, for example: "Welcome home! I've warmed up the living room and started your playlist." This response reflects real-time state validation and synthesis of multimodal information into coherent interaction. In general, this workflow demonstrates the multimodal reasoning integration, agent-based task coordination, and unified device control through MCP of HARMONY, allowing for scalable, adaptive, and intelligent ambient computing in smart homes.

Fig. 4 presents the layered architecture, illustrating the end-to-end data flow from multimodal input capture (Layer 1) through MLLM processing (Layer 2), agent dispatch (Layer 3), MCP communication (Layer 4), to device actuation (Layer 5). This five-layer structure enables seamless integration of cross-device coordination, intelligent task delegation, and multimodal perception for proactive, personalized smart home experiences. These agents communicate with smart home devices via a standardized MCP layer (Layer 4), enabling state queries, action execution, and event subscriptions. Finally, Layer 5 shows integration with diverse Samsung smart devices. Reactive automation triggered by explicit user commands, with potential for environmental context integration. The architecture consists of five strongly coupled layers, each of which is tasked with a key stage of perception, reasoning, and actuation.

At Layer 1 (User & Environment), the system receives a broad spectrum of multimodal sensor data from users and the environment, including gestures, visual scenes, speech and audio cues, and text commands. The various modalities give a rich representation of user context and intention and improve the semantic interpretation of interactions.

In Layer 2 (Input Processing & MLLM Core), the MLLM (for instance, Qwen 2.5 VL) does input fusion, integrates multiple streams of sensors into a single semantic description. It then does intent recognition to understand the user's high-level intentions, and task planning, in which executable actions are obtained. These actions are resolved to the corresponding AI agents in a terminal agent dispatch step.

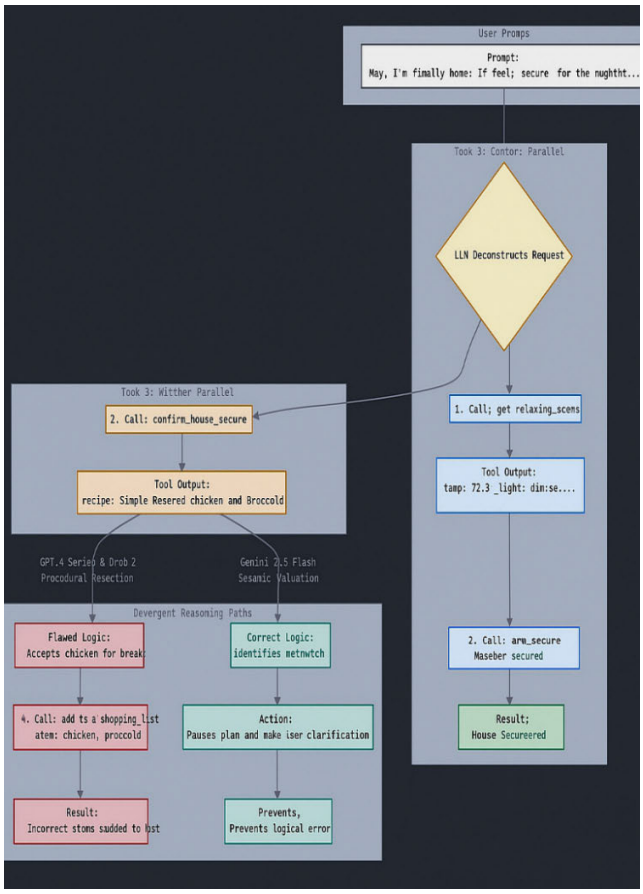

FIGURE 4. Parallel Multimodal Task Decomposition And Agent-Orchestrated Execution In A Smart Home Scenario Using Harmony Framework.

Layer 3 (AI Agent Layer) consists of domain-expert AI agents as detailed in Table 2. The Orchestrator Agent synchronizes cross-agent workflows, managing synchronization and conflict resolution in multi-agent systems. For example, the Comfort Agent manages environmental parameters such as lighting and Heating, Ventilation, and Air Conditioning (HVAC); the Entertainment Agent manages media playback and display settings; the Security Agent monitors surveillance and access control; the Kitchen Agent coordinates cooking operations and recipes; the Resource Management Agent (RMA) maximizes utility usage; and the Orchestrator Agent synchronizes cross-agent workflows, managing synchronization and conflict resolution in multi-agent systems.

These agents interact with the underlying device infrastructure via Layer 4 (MCP)—an extensible and standardized communication framework. AI agents may uniformly query and control diverse devices because of MCP's encapsulation of device-specific APIs. State representation (e.g., device_state({temp: 22, mode: 'cool'})), action invocation (e.g., set_temperature(device_id, temp: 24)), event

TABLE 1. Simulated MCP server configuration and functionalities for smart home evaluation.

Server Name	Simulated Device	Port	Key Functionalities
Samsung-WindFree-AC	Samsung WindFree™ Avant AC	8001	Power control, temperature/mode settings, AI auto comfort, energy monitoring via SmartThings.
Samsung-Smart-TV	Samsung 85" Neo QLED 8K TV	8002	Power/volume/input control, app launching, SmartThings integration for notifications.
Samsung-AILaundry-Combo	Bespoke AI Laundry Combo™	8003	AI cycle recommendations, progress reporting through SmartThings app.
AeotecMultiSensor6	Aeotec MultiSensor 6	8004	Contact state, temperature, battery reporting; motion detection.
Samsung-Refrigerator	Bespoke AI Family Hub™+ Fridge	8005	Inventory management, recipe suggestions, shopping lists via AI app.

TABLE 2. HARMONY AI agent roles and responsibilities.

Agent Name	Primary Responsibility	Key Samsung Devices Interacted With (Examples via MCP)	Example Multimodal Triggers / Inputs Processed with Multimodal Large Language Model Controller (MLLM-C)
Comfort & Ambiance Agent (CAA)	Manages climate, lighting, window coverings, background audio	Samsung WindFree™ AC, Smart Blinds, Philips Hue/SmartThings lights, Samsung Soundbars	- User says "It feels a bit stuffy and dark in here." (Audio + MLLM infers location) - Visual: Sunlight glare on TV screen detected by camera.
Security & Safety Agent (SSA)	Monitors for threats, manages access, ensures safety	SmartThings Cams, Door/Window Sensors, Samsung Smart Lock, Smoke/CO Detectors, Bespoke Jet Bot AI+™ (for patrol/anomaly sound detection)	- Audio: Glass breaking sound. - Visual: Unrecognized person at the door via SmartThings Cam. - Text: "Lock all doors when I leave."
Kitchen & Culinary Agent (KCA)	Assists with cooking, manages food inventory, kitchen tasks	Bespoke AI Oven/Microwave, Bespoke Family Hub™+ Refrigerator, Bespoke AI Dishwasher, Samsung Smart Cooktops	- Visual: User shows ingredients to camera + Audio: "What can I make with these?" - Audio: "Preheat oven to 350 for cookies." - Visual: Pot boiling over.
Entertainment & Well-being Agent (EWA)	Manages media playback, content discovery, wellness routines	Samsung Neo QLED/The Frame TVs, The Freestyle Projector, Galaxy Watch (wellness data), Samsung Soundbars, Smart Speakers	- User points at TV + Audio: "Play the next episode." - Visual: User looks stressed (facial recognition) + Galaxy Watch high HR data.
Resource Management Agent (RMA)	Optimizes energy/water usage, manages appliance schedules	SmartThings Energy service, Bespoke AI Washer/Dryer, Bespoke AI Dishwasher, EV Chargers	- Text: "Run dishwasher on cheapest cycle tonight." - Visual: Lights on in empty room via camera.
Chore & Maintenance Agent (CMA)	Manages cleaning schedules, device maintenance reminders	Bespoke Jet Bot AI+™ Robot Vacuum, Bespoke AI Washer/Dryer (lint filter), Family Hub+ (water filter), Air Purifier (filter)	- Audio: "When was last robot vacuum run?" - Visual: Dust detected by MLLM camera. - Text: "Remind me to clean AC filter."
Orchestrator Agent (OA)	Coordinates agents, resolves conflicts, manages system goals	Indirectly all devices via other agents; monitors system state	- Multi-step requests across domains. - Learns user preferences for agent priorities.

subscription (e.g., subscribe('door_opened')), and information querying (e.g., get_fridge_contents()) are some of its primary features. Because MCP is based on a client-server architecture, it is both modular and reusable across systems. AI agents serve as MCP clients, issuing requests, and smart home appliances make their capabilities available as MCP servers.

Lastly, Layer 5 (Samsung Smart Devices) includes the physical IoT ecosystem within the HARMONY system. Devices like the Bespoke Family Hub™, Neo QLED TV, Bespoke AI Washer, WindFree™AC, SmartThings Cam, Smart Lock, Smart Lighting, and Galaxy Devices have MCP interfaces. By using the MCP interface, the AI agents are able to invoke the high-level functionalities of the devices (e.g., temperature, playing media, biometric authentication, and camera streams) to activate them correctly and reliably. This multi-layered structure is a state-of-the-art integration of cross-device unity, smart job delegation, and multimodal AI perception. By abstracting away complexity, interoperability, and continuity of context across devices and situations, HARMONY provides proactive, responsive, and highly personalized smart home experiences.

IV. RESULTS-CASE STUDY: SIMULATION AND EVALUATION

This section presents a case study illustrating the operation and performance of the proposed HARMONY framework within a fully simulated smart home environment. All results presented are derived from simulation and do not represent real-world deployment.

A. FRAMEWORK

The simulation offers a structured context for evaluating the system's capability to process multimodal, context-aware inputs and coordinate intelligent responses through specialized AI agents and device-level actions via the MCP. MCP is a modular client–server framework that standardizes how AI applications connect to external tools, data sources, and services via MCP Hosts, MCP Clients, and lightweight MCP Servers. The servers communicate with local or remote data sources, offering one "AI USB-C"-a semantic interface enabling agent-tool integration across the ecosystem in a unified way. Samsung SmartThings exemplifies multimodal AI in consumer technology by integrating ambient sensing, voice recognition, and context-aware automation for personalized smart home experiences.

SmartThings Ambient Sensing is a key component of this infrastructure through the utilization of distributed motion and audio sensors to make inferences on humans' presence and activities like cooking, exercise, or work. This allows the system to actively shape and adjust the environment in real-time—remapping lighting, heat, and even behavioral nudges, and thus standing reminders for non-active users. Household appliances like the Family Hub refrigerator push this intelligence further with vision-based grocery recognition via the AI Vision Inside™camera, which is able to identify over 30 different grocery categories and offer personalized recipes. In addition, the SmartThings Hub integrated within the fridge serves as a primary command node to provide multimodal AI agents with the capability to coordinate interactions with other home devices.

To validate the architecture, the system was experimentally tested on a two-phase simulated home environment. In Phase 1, a high-fidelity testbed was built with five independent MCP servers coded in Python. Samsung smart device system includes several devices with distinct functionalities as shown in the Table.1: the Samsung WindFree™Avant AC controls power, temperature, modes, and monitors energy via SmartThings; the Samsung 85'' Neo QLED 8K TV manages power, volume, inputs, app launching, and SmartThings notifications; the Bespoke AI Laundry Combo offers AI-driven cycle recommendations and progress reporting; the Aeotec MultiSensor 6 detects motion and reports contact state, temperature, and battery status; and the Bespoke AI Family Hub™ + Fridge handles inventory management, recipe suggestions, and shopping lists through an AI app. Table 2 outlines the roles and responsibilities of Harmony AI Agents, each designed to manage specific smart home functions by interacting with Samsung devices and processing multimodal inputs through a Large Language Model Controller. These agents handle tasks ranging from comfort and security to kitchen assistance, entertainment, resource management, maintenance, and overall system coordination. Every server emulated a Samsung smart device running on individual ports with HTTP-based Server-Sent Events (SSE). The servers simulated dynamic state transitions such as power usage and motion, mimicking real-world behavior in a stateful, asynchronous way. Phase 2 consists of testing six cutting-edge large language models (LLMs): GPT-4.1, GPT-4.1-mini, GPT-4o, GPT-4o-mini (from OpenAI), Gemini 2.5 Flash (Google), and Grok 2 (xAI). The six models were deployed in a simulated smart home environment and tasked with executing four complex prompts.

The sequence that follows shows how HARMONY's LLM-driven orchestration provides context-aware, tailored smart home automation by utilizing multimodal input, real-time inference, agent-based task delegation, and device-level actuation via MCP. The Multimodal Large Language Model Controller (MLLM-C) analyzes natural language input, visual inputs (e.g., user gestures or room state), and environmental sensor data to glean high-level intent. The prompt is broken down into parallel sub-intents within three functional domains—comfort optimization, culinary preparation, and security assurance—each addressed by specialized AI agents through the MCP.

1) MULTIMODAL INTERACTION INITIATION

The user engages the system through a natural, multimodal command—by pointing at a Samsung The Frame TV displaying a vibrant artwork and saying, "I like this style, find something similar for the evening and dim the lights a bit." This triggers a coordinated input capture: a nearby

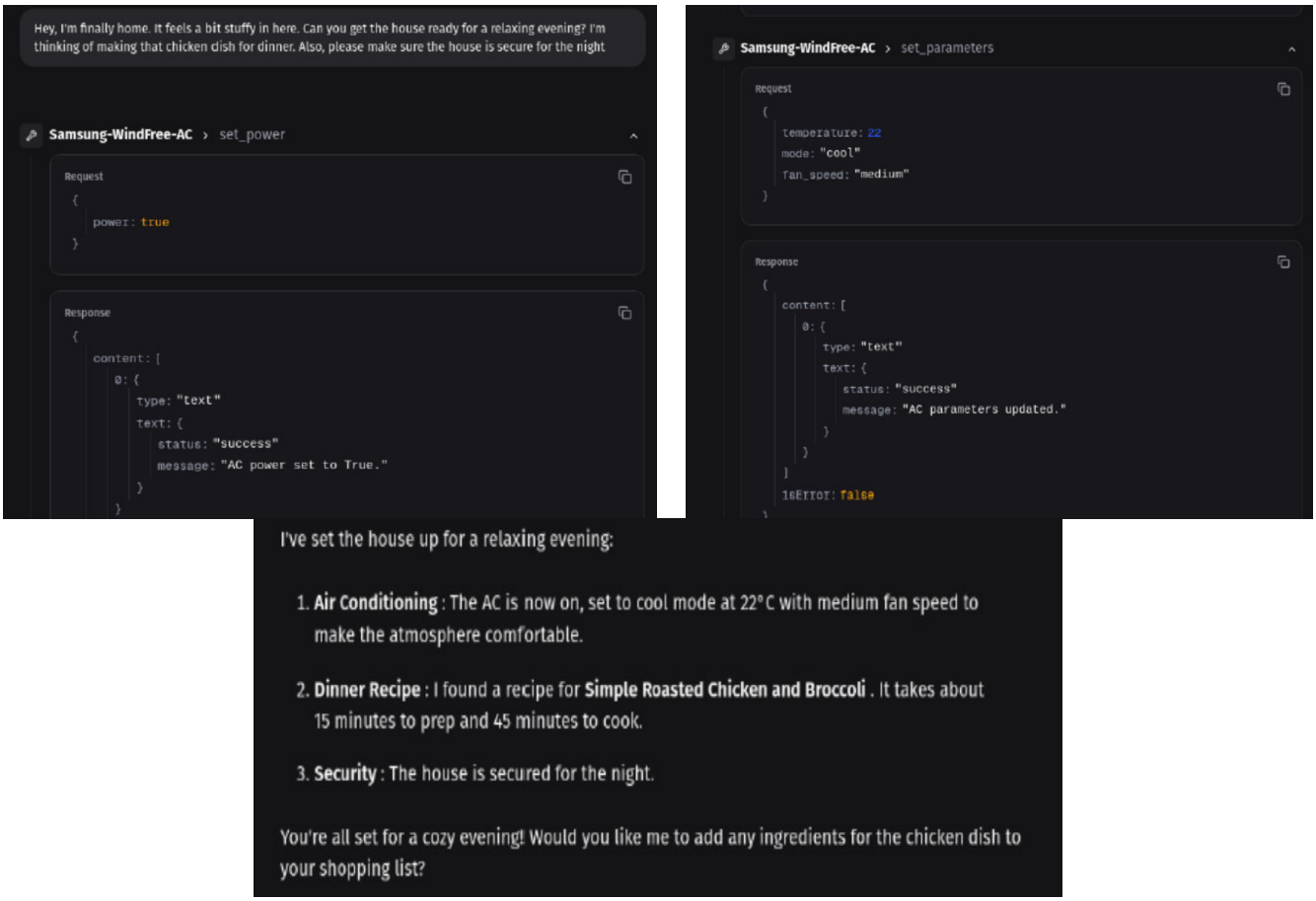

FIGURE 5. **Illustrative output from prompt-1 showcasing GPT-4o Mini orchestrating a comfortable evening setup.**

SmartThings Cam or Galaxy phone records the gesture and visual scene, while an onboard microphone captures the voice command, forming a rich contextual data stream.

2) INPUT FUSION AND MLLM CORE REASONING

MLLM-C processes and fuses the multimodal inputs. It transcribes the speech, analyzes the gesture, and identifies the art style currently shown on The Frame TV. Through contextual synthesis, the system infers the user's intent: to locate similar art for scheduled display later in the day and adjust the lighting for a more relaxed ambiance.

3) INTENT DECOMPOSITION AND AI AGENT DELEGATION

The coordinating Orchestrator Agent decomposes the user's high-level request into discrete tasks and dispatches them to the appropriate specialized AI agents:

Task 1: Locate artwork similar in style to the current display assigned to the Entertainment & Well-being Agent (EWA).
Task 2: Schedule the selected art for evening display which is handled by EWA.
Task 3: Adjust lighting levels for comfort as directed to the Comfort & Ambiance Agent (CAA).

4) EXECUTION THROUGH MCP-ENABLED DEVICE CONTROL

Each AI agent interfaces with smart devices via the MCP to perform their respective actions:

- **EWA Agent** queries The Frame using get_current_art_details(), leverages the MLLM-C's visual analysis (or an external art API) to identify stylistically similar pieces, and sends schedule_art_display(art_id, time='evening') to queue the artwork.

CAA Agent invokes get_light_status(group='living_room') to retrieve current lighting parameters, computes an appropriate dimming adjustment (e.g., 20% reduction), and dispatches the corresponding command via MCP.

B. USER FEEDBACK AND MULTIMODAL CONFIRMATION

Finally, the MLLM-C synthesizes the outcomes and delivers a natural language confirmation via a Galaxy device or Family Hub+ screen.

"Okay, I've found a Monet-style piece and scheduled it for display this evening. The living room lights have also been dimmed for a more relaxed setting." This ensures the user remains informed and reassured, maintaining a seamless ambient intelligence experience.

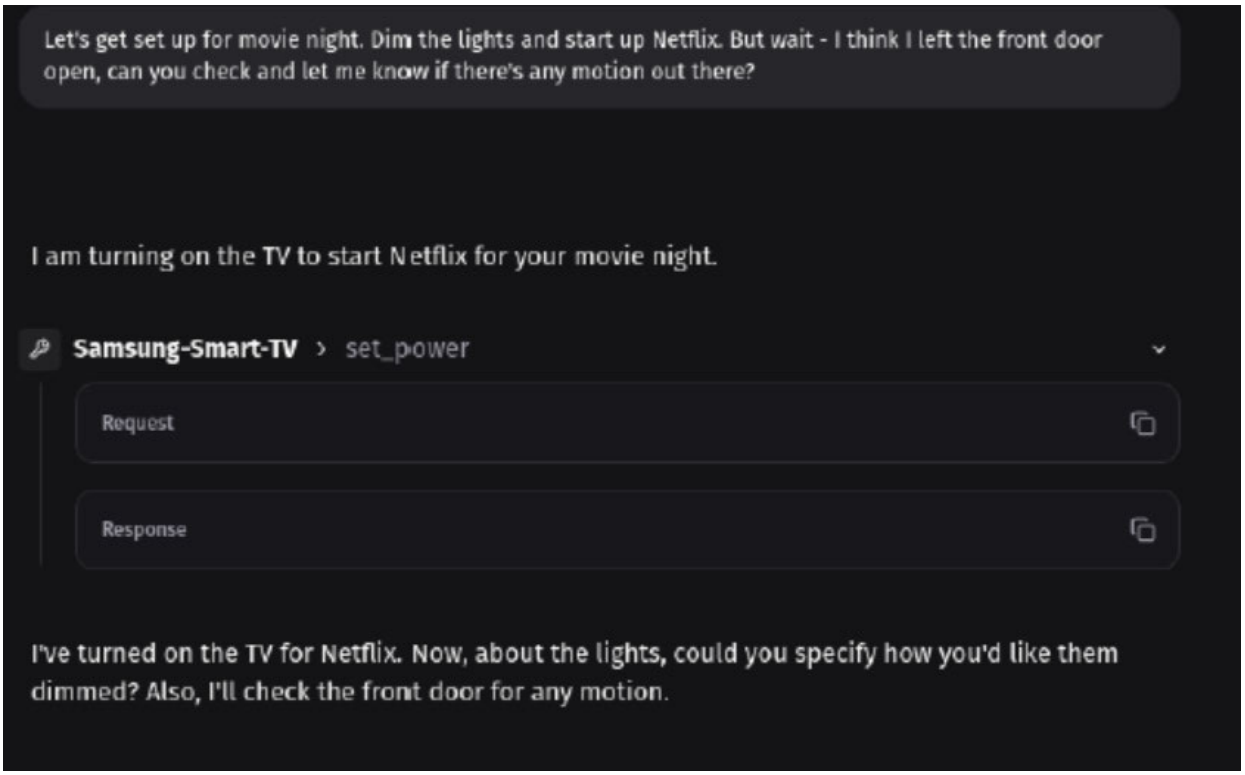

FIGURE 6. Example response from prompt-2 demonstrating grok 2 initiating a smart home setup for movie night.

a: GRANULAR PARAMETERIZED ACTION CONTROL

Every agent sends specific, parameter-laden commands to devices. The CAA, for example, does more than just turn on the air conditioner; it also sets the temperature to a precise amount (set_temperature: 22°C) and modifies the brightness of the lights to a predetermined level (adjust_brightness: -20%). Similarly, in convection_bake mode, the Kitchen Agent may send a complex order to preheat the oven to 200°C. To guarantee that AI replies closely match user preferences and contextual subtleties, this fine-grained control is essential.

b: SEMANTIC DEVICE STATE REPRESENTATION

Devices provide the LLM with status data in a format that can be understood semantically through the state representation. Smart gadgets like the Bespoke AI Oven provide structured, interpretable state descriptions like "current_mode": "convection_bake", "target_temp": 200, or "door_status": "closed" in place of low-level telemetry (e.g., "valve_state": 0). Without requiring device-specific hardcoding, this abstraction allows agents to reason about device states and change their behavior.

c: DYNAMIC CAPABILITY DISCOVERY

During runtime, agents can use MCP to query a device's supported features. For example, the CAA can query, "What lighting scenes are supported in the living room?" or the Security Agent might ask, "Does this door lock auto-lock?" By enabling agents to adapt to changes in firmware and device variances, this discovery strategy promotes ecosystem resilience and scalability.

d: ADVANCED MULTIMODAL INFORMATION RETRIEVAL

For example, the Kitchen Agent uses advanced perception pipelines to correlate expiration dates with ingredient metadata and retrieve inventory information from the Family Hub+ refrigerator's internal camera. Based on this, it identifies the "chicken dish" that the user has said, suggests a recipe that goes with it, and starts the cooking process. This demonstrates how HARMONY may link language, vision, and structured data to support practical decision-making.

e: EVENT-DRIVEN AGENT INTERACTION

Devices that generate status updates (such as "oven preheat complete," "motion detected in hallway," or "laundry cycle finished") are part of the event-driven system. In response to these occurrences, agents proactively subscribe. For example, the system exhibits predictive behavior based on real-time event handling when it detects the user's presence via a motion sensor and starts a series of ambient modifications before the voice command is entirely processed.

f: ACCESS AND SECURITY GOVERNANCE

Every agent has scoped access; for example, the Security Agent can arm and disable alarms, while the Kitchen Agent

cannot view video streams. Additionally, TLS-based encryption safeguards all inter-process communications, avoiding sensitive device orders and private user data from being accessed or intercepted by unwanted parties.

C. USE CASE ANALYSIS

The use case analysis discusses about how leading LLMs—GPT-4o Mini, GPT-4.1, GPT-4.1 Mini, GPT-4o, Gemini 2.5 Flash, and Grok 2—perform orchestration and reasoning tasks in smart home environments, using two compound user prompts designed to test real-time context interpretation, task decomposition, agent dispatch, and semantic reasoning. There are three use cases for understanding the HARMONY orchestration:

Prompt 1: Simulates a typical post-arrival smart home interaction, where the user returns home and issues a multi-intent command encompassing comfort, culinary preparation, and nighttime security. The natural language input—

"Hey, I'm finally home. It feels a bit stuffy in here. Can you get the house ready for a relaxing evening? I'm thinking of making that chicken dish for dinner. Also, please make sure the house is secure for the night."

—tests an LLM's ability to interpret nuanced human intent and coordinate multiple subsystems (HVAC, lighting, kitchen devices, and security tools) in parallel. It evaluates the system's contextual understanding, task decomposition, agent dispatch, and preference recall across environmental and procedural domains.

This multi-purpose prompt addresses four task categories:

- **Environmental Control:** Ease discomfort—tweak HV AC or clean the air.
- **Ambience Setup:** Dim lighting, play soothing music, or turn on scent diffusers.
- **Kitchen Assistance:** Fetch or offer the ''chicken dish'' and initialize proper appliances.
- **Home Security:** Secure doors, turn on surveillance, and verify safety measures.

Fig. 5 illustrates how the HARMONY system interprets a compound user command involving ambiance adjustment, culinary preparation, and home security. The MLLM core parses the natural language input, identifies multimodal cues, and decomposes the intent into parallel subtasks. These are then delegated to specialized AI agents—such as CAA, Kitchen Agent, and Security Agent—each interfacing with smart home devices via the MCP. The result is a synchronized, context-aware response that reflects personalized, semantically grounded automation. A parallel multimodal task orchestration workflow enabled by the HARMONY framework, showcasing how a compound user prompt—*"Hey, I'm finally home. It feels a bit stuffy in here. Can you get the house ready for a relaxing evening? I'm thinking of making that chicken dish for dinner. Also, please make sure the house is secure for the night."*—is semantically parsed and executed by an LLM-driven smart home system.

Prompt 2: Prompt 2 introduces a home entertainment scenario with a dynamic mid-command shift:

"Let's get set up for movie night. Dim the lights and start up Netflix. But wait – I think I left the front door open, can you check and let me know if there's any motion out there?"

This situation stacks:

- **Ambience Tasks:** Dimming lighting and Netflix boot-up.
- **Security Check:** Door status verification and motion detection.
- **Interrupt Handling:** Sudden change of user focus, requiring real-time context switching and conditional logic.

Fig. 6 shows a smart home AI interface responding to a user's compound voice command. The system confirms turning on the TV for Netflix, asks for lighting preferences to dim the lights, and indicates it will check the front door for motion, demonstrating multimodal task handling and coordinated smart device control. This tests an LLM's ability to sequence events, respond to interruptions, and pivot focus from routine ambient tasks (lighting, media control) to reactive safety concerns (door status, motion detection). It emphasizes contextual switching, state monitoring, and multi-agent collaboration, and provides insight into how well models balance proactive execution with reactive situational awareness.

The summarized model-to-model comparative evaluation for the prompts 1 and 2 is as follows:

1) GPT-4O MINI

- **Prompt 1**: Executes simple HVAC changes but misses multi-domain coordination; operates linearly and lacks task parallelism.
- **Prompt 2:** Handles light dimming but skips or simplifies security checks. Fails to maintain session continuity across user interruption.
- **Inference:** Best for lightweight automation. Lacks agentic behavior and contextual depth.

2) GPT-4.1 MINI

- **Prompt 1:** Segments tasks more cleanly, toggling HVAC, triggering known recipes, and activating door locks in sequence.
- **Prompt 2:** Recognizes the user's shift from ambience to security. Makes API-like calls to check door status and motion.
- **Inference:** Demonstrates structured orchestration and moderate contextual awareness. Suitable for smart home routines with limited ambiguity.

3) GPT-4.1

- **Prompt 1:** Offers clear decomposition, supports task chaining across HVAC, kitchen, and security modules.
- **Prompt 2:** Introduces device-specific reasoning (e.g., checking front door sensor logs) and conditional feedback loops.

- **Inference:** Robust orchestration for deterministic, event-driven environments. Begins to exhibit basic feedback generation.

3) GPT-4O

- **Prompt 1:** Executes HVAC and ambiance in parallel. Recalls "chicken dish" via memory grounding, while handling environmental state.
- **Prompt 2:** Combines motion data and door status, delivering feedback like "door is open, but no recent motion detected." Demonstrates asynchronous operations.
- **Inference:** A multimodal-ready orchestrator with strong API handling. Lacks high-level validation but excels in parallelism and real-time responsiveness.

4) GEMINI 2.5 FLASH

- **Prompt 1:** Pauses execution to validate user intent—e.g., verifies appropriateness of chicken dish for dinner. Proceeds only after clarification.
- **Prompt 2:** Offers a confidence-rated security report ("motion detected 2 mins ago"), then requests user confirmation before locking doors.
- **Inference:** Operates like a cautious orchestrator. Prioritizes semantic alignment and safety. Ideal for environments demanding trust, correction, and logical fidelity.

5) GROK 2

- **Prompt 1:** Executes singular commands (e.g., adjusting HVAC), but ignores compound intent and inter-agent workflows.
- **Prompt 2:** Dims lights and launches Netflix but omits or mismanages the security request entirely.
- **Inference:** Functions as a basic parser. Lacks state awareness, coordination, and decision branching making less suitable for complex orchestration.

Prompt 3:

The following prompt tests the ability of LLMs to pick up emotional subtext and react with empathy-based, health-promoting suggestions. It aims to gauge how effectively the models are able to make inferences about user requirements based on affective signals and take context-relevant wellness cues. This prompt models realistic interactions where users do not directly articulate their desire but portray mood or unease, letting the system "read between the lines" to respond appropriately. This prompt measures the model's performance in emotional sensitivity, semantic reasoning, and aligned assistance, especially valuable in wellness and assistive living domains.

"I've been feeling kind of off today…a bit tired and sluggish. Maybe I should try something healthier tonight."

This prompt measures the model's performance in emotional sensitivity, semantic reasoning, and aligned assistance, especially valuable in wellness and assistive living domains. The scenario allows for an optional follow-up confirmation if the model probes for clarification. Models are evaluated on their ability to:

- Recognize the implied intent (e.g., dietary adjustment, enhancing ambient environment),
- Offer suitable suggestions (e.g., recipes, lighting changes, relaxation cues),
- Engaging in gentle clarification to the user while avoiding overstepping or inappropriate assumptions strikes the perfect balance for user trust and effective AI interactions.

TABLE 3. Sample output of the prompt-3.

Model	Tool Calls	Response
GPT-4o-mini	set_power(TV), set_power(AC)	Turns on TV and AC for relaxation.
GPT-4.1	set_power(AC), set_parameters(AC), set_power(TV), set_volume(TV)	AC to 22°C WindFree, TV on volume 15.
GPT-4.1-mini	set_power(AC), set_parameters(AC), set_power(TV), set_volume(TV)	AC to 24°C cool low fan, TV volume 20.
GPT-4o	set_parameters(AC), set_power(TV)	AC 22°C low fan, TV on for comfort.
Gemini 2.5 Flash	None	Sympathy for allergies, no actions.
Grok 2	set_parameters(AC)	AC WindFree Cool 22°C low fan for allergies.

Table 3 presents sample outputs from different AI models responding to prompt commands, showcasing their tool calls and corresponding actions for controlling smart home devices like TVs and air conditioners. GPT-4o Mini performs basic, linear actions like HVAC control or media launching, but lacks multi-domain reasoning and fails to manage contextual transitions, making it suitable only for simple automation. GPT-4.1 Mini shows improved task segmentation and can respond to prompt interruptions with moderate orchestration and basic API-like interactions, but it still lacks deeper semantic flexibility. GPT-4.1 exhibits robust multi-step planning, integrates device-specific logic, and provides conditional feedback—ideal for structured, event-driven environments. GPT-4o excels in parallel task execution, memory-grounded actions, and asynchronous coordination (e.g., combining motion and door state data), offering high responsiveness though with limited semantic validation. Gemini 2.5 Flash stands out for its cautious, context-sensitive execution. It validates intent before acting

(e.g., pausing to confirm the suitability of a chicken dish), produces confidence-rated feedback (e.g., motion detected with timestamps), and prioritizes user alignment and safety, making it optimal for trust-critical applications. In contrast, Grok 2 lags significantly, only performing basic one-shot tasks with no coordination or context handling, unsuitable for complex orchestration. In Prompt 3, which assesses empathy and health-related context, GPT-4 series models generated actionable, comfort-enhancing commands (e.g., adjusting AC and TV settings), while Gemini 2.5 Flash offered only empathetic verbal support without action. Grok 2 managed minimal physical actions effectively but faced challenges in extracting nuanced emotional understanding. Overall, the results demonstrate a clear evolution from simple command execution to multi-agent orchestration that is cognizant of context, with Gemini exhibiting the most careful and semantically aligned behavior.

D. EVALUATION FAIRNESS AND RESOURCE DISCLOSURE

This section elucidates a fair comparison between commercial LLMs and potential open-source models by ensuring fair evaluation of different LLMs by standardizing experimental conditions and disclosing the computing resources available to each model.

1) TO MAINTAIN FAIRNESS, ALL MODELS WERE TESTED UNDER IDENTICAL SETTINGS

- **Sampling parameters:** temperature 0.7, top-p 0.9, max tokens 2048
- **Unified MCP setup:** same JSON-RPC 2.0 tool definitions
- **Network simulation:** fixed 20ms ±5ms latency
- **Deterministic device states:** replayed logs with random seed 42.

These controls ensure differences arise from model capability, not test environment variations.

2) COMPUTE RESOURCES FOR EACH MODEL

The compute resources for each model play a major role in shaping its performance, cost, and suitability for real-world deployment. Commercial API models such as GPT-4.1, GPT-4o, Gemini 2.5 Flash, and Grok 2 run on large proprietary cloud infrastructures with advanced hardware (A100/H100-class GPUs or TPUs) and highly optimized inference pipelines. This gives them strong reasoning capabilities, stable latency, large context windows, and mature safety tuning—but also introduces higher operating costs and external rate limits.

Table 4 summarizes the compute resources and performance metrics for various AI models, detailing their infrastructure, context window sizes, average input/output token counts, latency, and rate limits. In contrast, open-source models like LLaMA 3.1/3.2 can be deployed locally on user-controlled hardware. Although local deployment reduces the dependency on cloud services, lowers per-evaluation cost, and enhances privacy, the achievable performance is constrained by available GPUs, user-side optimization, and the model's own parameter size. These factors influence latency, context handling, and reasoning depth.

TABLE 4. Compute resources for each model.

Model	Infrastructure / Hardware	Context Window	Avg. Tokens (Input / Output)	Latency	Rate Limits
GPT-4.1 / GPT-4o	Proprietary cloud (A100/H100 GPUs)	128K (GPT-4o); 1M (GPT-4.1)	1,847 / 412	1.5–4.5s (cloud)	10 req/min
GPT-4.1 Mini	Optimized cloud accelerators	1M	1,652 / 287	Faster than GPT-4.1	10 req/min
Gemini 2.5 Flash	Google TPU v5e backend	1M	1,923 / 531	Similar to GPT-4	15 req/min
Grok 2	Custom GPU cluster (undisclosed)	32K	1,456 / 203	Not specified	5 req/min

E. QUANTITATIVE ANALYSIS

Each model was tested with 20 independent runs per prompt under controlled settings: fixed temperature (0.7), maximum tokens (2048), identical MCP schemas, and deterministic device state replay from time-stamped simulator logs. The following metrics were considered for quantitative analysis.

1. **Task Completion Rate (TCR)** – how often the model actually finished all required subtasks.
2. **Command Accuracy (CA)** – how precisely it issued the correct tool calls with the right parameters.
3. **Latency** – how long it took from perceiving a prompt to receiving the device's acknowledgment, broken into stages.
4. **Interruption Handling Success (IHS)** – whether the model could deal with a sudden "Wait, change that" without forgetting previous tasks.
5. **Safety Validation Rate (SVR)** – whether it asked the user for confirmation before performing sensitive actions like locking doors or activating cameras.

Table 5 shows the performance metrics and latency analysis of our simulated environment, demonstrating how the models tradeoff between execution speed, semantic reasoning, and agentic autonomy.The GPT 4.1 model is the

TABLE 5. Performance metrics and latency analysis.

Model	TCR (%)	CA (%)	IHS (%)	SVR	Latency Mean	p95 Latency
GPT-4.1	94.2	91.8	89	45	4.1	6.2
GPT-4o	82.1	78.3	71.5	38.5	3.5	5.4
Gemini 2.5 Flash	88.5	86.2	92	87.5	5.4	7.8
GPT-4o Mini	75.3	72.1	64	31	2.9	4.0
GPT-4.1 Mini	82.1	78.3	71.5	38.5	3.5	5.4
Grok 2	42.3	39.7	28.5	12	4.7	7.1

strongest performer with a task completion rate of 94.2% and a command accuracy of 91.8%. This is well supported by a moderate median response time of 3.8 seconds, reliably completing almost every assigned task and giving commands with high precision. The GPT 4o model emphasizes speed of 2.9-second median and a 4.2-second p95 latency, though it still maintains solid levels of accuracy, and task completion. This makes it an excellent choice in scenarios where rapid reaction time is crucial. Meanwhile, Gemini 2.5 Flash, though often prompting for confirmation before carrying out a task, channels this caution into thoughtful, reliable operation—an advantage when precision and deliberation are key. The Mini variants embody the classic strengths of compact models: they respond quickly, run efficiently, and deliver solid performance, even if they occasionally falter on complex or multi-step instructions. Their 75–82% completion rates reflect these manageable challenges. GPT Mini 4.0 distinguishes itself through its rapid response times—2.9 seconds median and 4.2 seconds p95—paired with high accuracy, consistent task reliability, and prudent confirmation for sensitive operations. Grok 2 lays a strong, promising foundation that holds tremendous potential for further refinement. With its reliable 42.3% completion rate and solid performance on simpler instructions and straightforward tasks effectively. Across all models, the shift from reasoning to device-control actions remains impressively swift, averaging 300–600 ms for internal handoff and an additional 200–400 ms for device acknowledgment—timings that reflect only minor simulator network and device delays.

From the above quantitative analysis, it is analyzed that,

- GPT-4.1 delivers the highest reliability and accuracy, while GPT-4o and the Mini variants prioritize speed over deeper reasoning and task completion. Gemini 2.5 Flash excels in safety and interruption handling.
- While analyzing the impact of safety and model size on latency, the models with stronger safety behavior (e.g., Gemini 2.5 Flash) show higher latency due to confirmation steps, whereas compact models respond faster but struggle with complex or multi-step tasks. All systems share similar device-side delays, with 300–600 ms for tool-call handoff and 200–400 ms for device acknowledgment.

F. VALIDATION ANALYSIS

For validating the effectiveness of the LLM orchestration, the Archetype matrix is used. This matrix categorizes state-of-the-art large language models (LLMs) based on their observed levels of proactivity (vertical axis) and contextual reasoning ability (horizontal axis) during smart home orchestration tasks as shown in Fig. 7. The "LLM Orchestrator Archetype Matrix" is a systematic assessment of different large language models (LLMs) against two essential dimensions that determine their fitness for autonomous orchestration activities: Reasoning Capability (horizontal axis) and Proactivity (vertical axis). The visual representation is specifically applicable within the realm of LLM-powered agent systems, where the capacity of the model to reason out through intricate tasks and take actions without persistent solicitation is pivotal for utility in the real world.

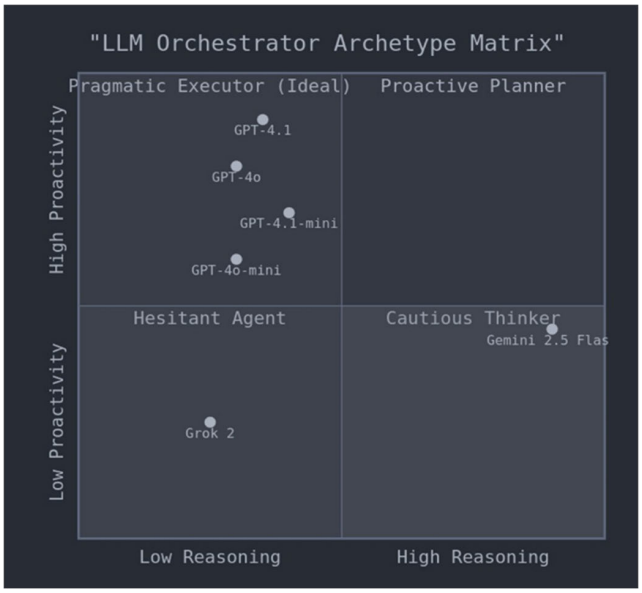

FIGURE 7. LLM orchestrator archetype matrix.

The matrix is split into four quadrants, each of which corresponds to a different archetype of LLM behavior:

1) PRAGMATIC EXECUTOR (HIGH PROACTIVITY, MODERATE TO HIGH REASONING)

This quadrant embodies the optimal balance for orchestrator agents—models that can rapidly parse goals, make bold

TABLE 6. Orchestration archetypes and insights.

Quadrant	Models	Characteristics
Pragmatic Executor	GPT-4.1, GPT-4o	High task throughput, low semantic friction, parallel-friendly
Cautious Thinker	Gemini 2.5 Flash	Validates semantics, aligns with intent, slows for accuracy
Hesitant Agent	Grok 2	Reasoning capabilities improve with structured, well-scoped prompts
Transitional Executors	GPT-4.1 Mini, GPT-4o Mini	Evolving multi-step planners; limited adaptation and contextual nuance

moves, and learn to fit context with appropriate reasoning accuracy. Models like GPT-4.1, GPT-4o, GPT-4.1-mini, and GPT-4o-mini fall into this region, evidence of their excellent real-time task execution capacity, effective context usage, and consistency in executing follow-through actions without excessive deliberation. GPT-4.1, positioned topmost in this quadrant, exhibits the maximum proactiveness in this group, making it extremely valuable for autonomous control flows in intelligent systems, Robotic Process Automation (RPA), and multimodal agents.

2) PROACTIVE PLANNER (HIGH PROACTIVITY, HIGH REASONING)

This is the theoretically ideal quadrant, in which a model would balance strong cognitive thought with the ability to plan and carry out multistep tasks. Notably, the existing models explores to fills this niche, and this represents a scope for developing robust architectural frameworks in existing LLM capabilities. This would allow models not only to respond to input effectively but also to actively create plans, handle dependencies, and adaptively re-plan within changing environments. This implies a potential for future development and research, most importantly reinforcement learning fine-tuning, agentic memory, and planning modules.

3) CAUTIOUS THINKER (HIGH REASONING, LOW PROACTIVITY)

Gemini 2.5 Flash is found to be prominent in the quadrant of high inferential and high analytical reasoning, which is best suited for decision trees, structured assessments, and formal knowledge reasoning, but slightly faces challenges in real-time orchestration that requires a high degree of rapid decisions with autonomous action. They are accurate and thoughtful but require more prompting to drive tasks effectively, making them more suited for assistant-style scenarios rather than fully autonomous roles.

4) HESITANT AGENT (LOW REASONING, LOW PROACTIVITY)

Grok 2 resides in the lower-left quadrant, embodying shallow reasoning and less proactive in nature when compared to other LLM models considered for this simulation. These models smartly remain passive with limited initiative, excelling in defined, and low-risk roles over autonomous orchestration. However, this matrix reflects only the current state of LLM-based orchestration capabilities. While some models particularly those in the GPT-4 lineage are increasingly designed for real-time, agentive operation. A still-emerging frontier lies in developing LLMs that can seamlessly integrate strategic foresight with autonomous action, which represents the true marker of advanced AI orchestration.

G. POTENTIAL IMPACTS OF INCORPORATING LLAMA MODELS

This section discusses whether the LLaMA models were incorporated into this evaluation, the impact on the analysis a few systematic differences is expected from architectural and deployment characteristics unique to open-weight models. First, task performance on text-dominated interactions would be highly competitive with leading commercial models. Benchmark evidence shows that LLaMA 3.1 instruct models demonstrate strong multi-step reasoning, robust instruction-following, and effective tool-use capabilities, indicating their ability to closely approach the performance of models such as GPT-4.1 and Gemini 2.5 on text-centric workflows. While multimodal orchestration presents a more challenging frontier, this primarily reflects architectural focus rather than fundamental limitations. LLaMA 3.1's text-first design allows it to excel in language-heavy reasoning tasks, and the availability of integrated vision extensions in the LLaMA 3.2 Vision family provides a clear pathway for multimodal expansion.

While current cross-modal reasoning and real-time perceptual inference may not yet match the polish of highly optimized commercial systems, these opportunities for growth are readily addressable through targeted enhancements in perception modules, streamlined infrastructure, and task-specific tuning. Overall, LLaMA's robust textual reasoning foundation, paired with its open and extensible ecosystem, establishes it as an exceptionally capable and versatile platform, ideally suited for applications prioritizing transparency, customization, and scalable optimization. In addition, LLaMA's deployability on local hardware could introduce different latency and resource trade-offs; while smaller LLaMA variants may outperform cloud APIs in raw response time, their reduced reasoning depth may affect multi-step orchestration quality. Integrating LLaMA models could create a performance profile with its own strengths and trade-offs. LLaMA models demonstrate impressive capabilities in text-based reasoning and offer robust privacy guarantees through on-device and customizable deployments,

making them especially well-suited for sensitive, regulated, or internal environments. It is observed that LLaMA models encounter challenges with multimodal robustness, safety alignment, and reliable orchestration of complex tool-based workflows. The positive attributes of LLaMA in terms of privacy, transparency, and logical reasoning make them a strong contender for specific use cases, and ongoing model advancements continue to address performance gaps in multimodal and orchestration domains.

V. LIMITATIONS OF PRESENT WORK

The framework is subjected to the following limitations Firstly, the MCP orchestration layer requires robust security measures including TLS encryption, scoped access tokens, sandboxed agent execution, and audit logging to prevent malicious endpoint exploitation.

Secondly, MCP integration across Matter, Zigbee, Wi-Fi devices faces computational and protocol compatibility challenges.

Third, advanced models like Gemini 2.5 Flash exhibit semantic awareness while maintaining fast response speeds suitable for many operations, though Pro variants may trade speed for deeper reasoning in complex scenarios. The dependancy on local or cloud infrastructure can impose scalability and latency constraints in deployments. Security and scalability challenges persist, yet the architecture templates future AI-orchestrated ecosystems.

Fourth, the framework relies on continuous multimodal data collection including audio, video, and user behavior patterns. This raises significant privacy concerns regarding data storage, transmission, and potential misuse. While MCP supports local processing, cloud-based LLM inference (e.g., GPT-4.1, Gemini 2.5) requires transmitting potentially sensitive contextual information to external servers. Future implementations is expected to focus on addressing data minimization, user consent mechanisms, and compliance with privacy regulations such as GDPR.

Fifth, the evaluation reveals median response latencies of 2.9-4.3 seconds across tested models. For time-critical applications such as security alerts or safety interventions, these latencies may be insufficient. Edge deployment of smaller models could mitigate this, but at the cost of reasoning depth. Achieving sub-second response times while maintaining semantic quality remains an open challenge for real-time smart home orchestration.

Sixth, the framework's orchestration quality depends heavily on commercial, closed-source LLMs (GPT-4.1, Gemini 2.5 Flash, Grok 2). This creates vendor lock-in risks, unpredictable API cost scaling, and limited transparency in model behavior. While Section IV-G discusses LLaMA integration potential, the current architecture lacks validated support for fully open-source orchestration, limiting reproducibility and deployment flexibility in regulated environments.

Seventh, the simulation evaluated a limited device ecosystem (5 MCP servers, 6 agent types). Scaling to real-world deployments with dozens of heterogeneous devices, multiple concurrent users, and cross-home coordination introduces challenges in agent synchronization, MCP message throughput, and conflict resolution that remain unexplored in this study.

To strengthen the framework beyond simulations, there is a potential to build a small real-world prototype. This setup may use a Samsung SmartThings Hub alongside Samsung appliances like the WindFree™AC and Family Hub+ Refrigerator, plus Matter-certified devices such as Philips Hue bulbs and an Aqara Thread contact sensor. Orchestration will run on a Raspberry Pi 5 hosting MCP servers and an edge-optimized LLaMA 3.2 11B model for fast local inference. The prototype will use the same logging as simulations—timestamped MCP messages, agent traces, frequent device state snapshots, and network telemetry. There is a scope to explore the study differences between simulation and reality, failure causes, and interoperability across Matter, Thread, Zigbee, and Wi-Fi devices.

In future works, the research will focus on the design of secure, privacy-protecting MCP solutions that are specific to domestic contexts, with fine-grained control. Future work should prioritize Matter-secured protocols, lightweight MLLMs, adaptive agents, and explainability for autonomous multimodal homes. Improvements in lightweight, edge-deployable MLLMs may have the potential to overcome performance limitations on low-power devices to provide wider adoption. Strengthening the proactive planning and memory functions of AI agents will also be essential to move towards long-term, adaptive behavior from reactive automation. In addition, an adaptive learning framework will confidently support continuous personalization, strengthening user trust and enhancing system explainability as the technology evolves.

VI. CONCLUSION

The study presents HARMONY, a multimodal AI-driven smart home framework that integrates MLLMs, specialized AI agents, and the MCP to deliver intuitive, proactive, and context-aware home automation. Through an evaluation of various large language models—from task-driven models like GPT-4o Mini to semantically reflective agents like Gemini 2.5 Flash—the study reveals a clear progression in LLM orchestration capabilities. The simulation-based results demonstrate that MLLMs, when integrated with standardized communication protocols, can reliably manage device states in controlled test scenarios, interpret multimodal inputs, and coordinate comfort, security, and kitchen operations through effective agent collaboration. The proposed framework further enhances personalization, decision-making, and operational efficiency in smart homes by enabling robust real-time, multimodal reasoning and smooth action deployment. Insights from the orchestration matrix indicate that GPT-4 variants offer the most balanced performance for real-time orchestrator roles in agent-based systems. The findings also highlight that strong reasoning capabilities with proactivity and autonomy are essential for effective orchestration. While

Gemini 2.5 Flash shows promising potential among the other LLMs considered for this simulation, it would benefit from additional advances in autonomous planning to fully realize its capabilities for orchestrator-centric smart home deployments.

APPENDIX
GLOSSARY OF TERMS

Term	Explanation
Ambient Intelligence	Environments using sensors and AI to perceive context and act proactively without user input.
API (Application Programming Interface)	Interface enabling software to control devices and retrieve statuses reliably across systems.
Command Accuracy (CA)	Measures the proportion of user commands or instructions correctly interpreted and executed by a model, often in voice assistants, chatbots, or task-oriented systems
Comfort Agent (CAA)	A Comfort Agent in the context of SmartThings or smart home systems typically refers to an AI-powered feature or agent that learns user preferences and adjusts environmental settings—like temperature, lighting, or appliance controls—to enhance personal comfort automatically and seamlessly.
Context Awareness	Dynamic interpretation of user needs, environment, and history for adaptive responses.
Domain-Specific AI Agents	Specialized agents (e.g., Comfort, Kitchen, Security) executing tasks delegated by the MLLM and interacting with devices via MCP.
Dynamic Capability Discovery	Querying devices at runtime to determine their supported features, enabling flexible orchestration across diverse hardware.
End-to-End Multimodal Orchestration	Complete pipeline from multimodal perception to MLLM reasoning and device actuation via agents and MCP.
Event Subscription	Mechanism for receiving real-time updates from devices (e.g., motion detection, preheat complete), enabling proactive behavior.
Entertainment Agent (EWA)	An AI agent managing entertainment by integrating ambient sensing with media playback, lighting, and sound adjustments.
Interruption Handling Success (IHS)	Focuses on AI resilience to mid-response disruptions, common in voice assistants or real-time systems; success rate tracks seamless recovery or handling without failure.
LLM Orchestrator Archetype Matrix	Categorizes LLMs by reasoning ability and proactivity (e.g., Pragmatic Executor, Cautious Thinker).
MCP Client	AI agent communicating with MCP servers to send commands and retrieve data.
MCP Host	Application or process requiring access to device states, tools, or external data via MCP.
MCP Server	Lightweight module exposing device functions (state, actions, events) through the MCP interface.
Model Context Protocol (MCP)	Standard communication protocol enabling uniform querying, action invocation, and event handling across devices.
Multimodal Input Fusion	Combining audio, text, visual, and sensor data streams into a unified semantic representation for intent recognition.
Multimodal Large Language Model (MLLM)	AI model capable of interpreting and reasoning over text, audio, and visual inputs simultaneously
Open-Weight Models (LLaMA 3.x)	Locally deployable models that offer transparency and privacy but generally have weaker multimodal and orchestration performance.
Orchestrator Agent (OA)	Coordinates multi-agent workflows, resolves conflicts, and manages cross-domain tasks within HARMONY.
Safety Validation Rate (SVR)	Frequency with which a model requests confirmation before executing sensitive or safety-critical tasks
Sim-to-Real Gap	Differences between simulated and real-world performance caused by network variability, sensor noise, and hardware constraints.
Smart-Home Latency Domains	Time delays across sensing (400–800 ms), cloud inference (1.5–4.5 s), and device actuation (200–400 ms).
SmartThings Ecosystem	Samsung's platform for IoT device integration, sensing, and automation
State Representation	Structured and interpretable description of device status used for semantic reasoning.
Task Completion Rate (TCR)	Measures the percentage of assigned tasks successfully completed within a given timeframe, calculated as (Number of Completed Tasks / Total Number of Tasks) × 100

Tool Call / Action Invocation	API-like command sent through MCP to modify device behavior or query its state.

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